

ArrayList – Complete Notes

What is an ArrayList?

- ArrayList is part of the Java Collection Framework and implements the List interface.
- Elements are ordered using index values (0-based indexing).
- Internally uses a dynamic resizable array.
- Allows duplicates and null values.
- Default initial capacity = 10.

When to Use ArrayList?

- ✓ Fast retrieval – $O(1)$
- ✓ Good for frequent access operations

When NOT to Use ArrayList?

- ✗ Avoid when many insertions/deletions happen in the middle (shifting occurs – $O(n)$)

Important Methods

add(value), remove(index), get(index), set(index, value), clear(), subList(start, end), toArray(), trimToSize()

LinkedList – Complete Notes

What is LinkedList?

- Linear collection of nodes linked by pointers.
- Not stored in continuous memory.
- Fast insertion/deletion, slow retrieval.

Node Structure

Singly LL → Data + Next Pointer

Doubly LL → Data + Prev + Next Pointer

Features

- Supports duplicates & nulls
- Not synchronized
- Faster insertion/deletion
- Slow random access

LinkedList vs ArrayList Comparison

Index Support: LinkedList – No | ArrayList – Yes

Random Access: LinkedList – No | ArrayList – Yes

Most Expensive Op: LinkedList – Retrieval | ArrayList – Insert/Delete

Singly vs Doubly LL

Space: Singly – Less | Doubly – More

Implementation: Singly – Easy | Doubly – Hard

Insert/Delete Time: Singly – Less | Doubly – More

Navigation: Singly – One-way | Doubly – Two-way

Quiz Section

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1. Default size of LinkedList?

Answer: 0

2. Why doesn't LinkedList support random access?

Answer: It does not implement RandomAccess and requires traversal.

3. Worst-case insertion time in doubly LL?

Answer: $O(n)$ if traversal needed, $O(1)$ if reference known.

4. Method to trim ArrayList capacity?

Answer: trimToSize()